

ENVIRONMENTAL FACTORS INFLUENCING ODONATA COMMUNITIES OF THREE MEDITERRANEAN RIVERS: KEBIR-EAST, SEYBOUSE, AND RHUMEL WADIS, NORTHEASTERN ALGERIA

Amina YALLES SATHA^{1,2} & Boudjéma SAMRAOUI^{1,2,3}

¹ Laboratoire de Conservation des Zones Humides, University of Guelma, Guelma, Algeria. E-mails: bsamraoui@gmail.com & yallesamina@yahoo.fr

² University of 08 mai 1945, Guelma, Algeria

³ Biology Department, University of Annaba, Annaba, Algeria

RÉSUMÉ.— *Facteurs environnementaux influençant les communautés d'Odonates de trois rivières méditerranéennes : les oueds Kebir-Est, Seybouse et Rhumel, nord-est algérien.*— Les Odonates sont une composante importante des peuplements des milieux lotiques et leur abondance et diversité renseignent sur l'intégrité écologique de ces hydrosystèmes. L'inventaire odonatologique de trois oueds majeurs algériens : Kebir-Est, Seybouse et Rhumel, a permis l'identification de 40 espèces. Nos résultats révèlent la présence de *Calopteryx exul*, endémique maghrébin, dans l'oued Seybouse et semblent confirmer l'extinction de la population type dans l'oued Rhumel où l'espèce avait été découverte au XIXe siècle. Nos résultats indiquent également l'expansion de plusieurs espèces : *Coenagrion caeruleum*, *Orthetrum nitidulatum*, *Trithemis kirbyi* et *Urothemis edwardsii* dont la population relictuelle est en danger critique d'extinction. La mesure de diverses variables physicochimiques (altitude, température, conductivité, etc.) nous a permis d'explorer une possible co-structure entre les jeux de données faunistiques et de variables environnementales. L'analyse des données indique que la richesse spécifique est, selon l'oued, variablement corrélée à l'hydropériode, à la conductivité et à la température de l'eau, suggérant son utilité dans l'évaluation de l'intégrité écologique des cours d'eau méditerranéens.

SUMMARY.— Odonata are an important component of lotic communities and their abundance and diversity may inform on the health of running waters. The survey of the odonatofauna of three major Algerian wadis: Kebir-East, Seybouse, and Rhumel, led to the identification of 40 species. Our results revealed the presence of *Calopteryx exul*, a Maghrebian endemic, at Wadi Seybouse and seemed to confirm the extinction of the type population at Wadi Rhumel where the species was first recorded in the XIXth Century. The results also indicated the range expansion of several species: *Coenagrion caeruleum*, *Orthetrum nitidulatum*, *Trithemis kirbyi* and *Urothemis edwardsii* whose relict population is critically endangered. Additionally sampling of abiotic factors (altitude, water temperature, conductivity, etc.) was carried out and data analysed in view of exploring a possible co-structure between the faunistic and environmental data sets. Data analysis indicated that correlation between abiotic factors (hydropereod, water conductivity and water temperature) and species richness differed between streams and thus Odonata may prove useful in evaluating the ecological integrity of Mediterranean lotic systems.

Rivers and their floodplains are characterized by high habitat heterogeneity and biodiversity (Chovanec & Waringer, 2001), but are also among the most threatened ecosystems on Earth (Abell, 2002; Dudgeon *et al.*, 2006). These aquatic ecosystems are increasingly threatened particularly by dam construction, water abstraction, habitat modification, industrial effluents, domestic sewage, invasive species, and agricultural runoff including fertilizers and pesticides (Malmqvist & Rundle, 2002; Samways & Taylor, 2004).

North African wadis are generally intermittent rivers and streams that occur in a water-stressed region where the ever-increasing local needs for freshwater are exacerbated by the poor management of this precious resource in the context of climate change that is likely to widen the current water deficit. To educate the public and provide decision-makers and managers with tools they can use in management and conservation policies, there is an urgent need to develop technically easy and low-cost methods for rapid bioassessments and monitoring of the ecological integrity of Mediterranean lotic ecosystems (Oertli, 2008; Simaika & Samways, 2012).

Preliminary bioassessment using Odonata fauna of a number Algerian wadi was started with that aim in sight (Khelifa *et al.*, 2011; Benchalel & Samraoui, 2012; Bouchelouche *et al.*, 2015; Hamzaoui *et al.*, 2015; Hafiane *et al.*, 2016).

Odonata have been nicknamed “the Guardians of watersheds” (Clausnitzer & Jödicke, 2004; Clausnitzer *et al.*, 2012) and because Odonata communities have complex habitat requirements, their abundance and diversity reflect a healthy and stable environment (Castella, 1987; Clark & Samways, 1996). By occupying, as prey and predators a critical trophic place in both terrestrial and aquatic food webs, Odonata are particularly vulnerable to changes in their environment (Sahlèn & Ekestubbe, 2001). Their adult stages are sensitive to habitat structure which make them reliable sentinels of river disturbances (Clausnitzer, 2002) while their amphibious habits led them to be useful bioindicators of both land and freshwater ecosystems (Dijkstra & Lempert, 2003; Martín & Maynou, 2016).

The aim of our study was two-fold: (a) to map the distribution of each species and compare the odonatological communities of the three wadis, and (b) to identify how environmental factors are associated with the species richness of lotic Mediterranean Odonata.

MATERIALS AND METHODS

STUDY AREA

The Kebir-East watershed

The watershed covers an area of 3203 km². The sub-basin Oued Kebir-East extends from the Tunisian border in the east to the confluence of the Kebir-East and Bounamoussa wadis at the Mafragh. In the north, it is bounded by the coastal basin of El Kala and in the south by the sub-basin of coastal Bounamoussa (Fig.1 a,b). The Oued Kebir-East watershed occupies a large proportion of the coastal strip that span a distinct biogeographic zone known as eastern Numidia which extends from the Tunisian border to the lower reaches of Wadi Seybouse. The population hosted by the watershed is estimated at 308 811 inhabitants (ABHCSM, 1999).

The Kebir-East hydrosystem is 1780 km long and includes 35 wadis: Oued El Kebir, Bougous, Leben, Zitouna, Guergour, etc. The watershed is under the influence of humid and sub-humid Mediterranean climates characterized by a dry period from May to September and a wet season from October to April. The average annual rainfall varies from 750 mm to more than 1000 mm in Kroumiria.

The pollution within the Kebir-East watershed is mainly due to agriculture (excessive use of fertilizers and pesticides) and urban waste water (because most agglomerations are not connected to a sewage system). Industrial wastewater originates mainly from the food industry implanted at Bouteldja, Ben M'Hidi and Zrizer (Mebarki, 2009).

The Seybouse watershed

Wadi Seybouse is a Mediterranean river which originates in the semi-arid Hauts Plateaux (Fig. 1c). It drains a large basin with an area of 6570 km² and measures 134.7 km. The Seybouse watershed is 3000 km long and consists of 42 wadis which include Wadi Zenati, Wadi Bou Hamdane and Wadi Cherf. The latter two wadis unite at Medjaz Amar to form Wadi Seybouse which reaches the sea at Annaba (ABHCSM, 1999).

The climate of the watershed is typically Mediterranean and overlaps between two bioclimatic regions: the continental semi-arid climate of the Hauts Plateaux and the subtropical humid and sub-humid climates of Numidia. The average annual rainfall varies from 450 mm in the south to 750 mm in the north.

The water of Wadi Seybouse is mainly used for irrigation and industrial purposes. The Seybouse Basin discharges into the sea a volume of wastewater amounting to 500 ls⁻¹. The largest contribution comes from the coastal sub-basin which houses the majority of industries which include the steel complex of El Hadjar and the industrial zone of Pont Bouchet. Some effluents, from chemical factories like ASMIDAL and dairy plants, flow directly into the sea (Mebarki, 2009).

The Rhumel watershed

The Rhumel represents the largest watershed of Algeria and has a catchment area of 8800 km² with 2 056 000 inhabitants. It extends from the northern margin of the Hauts Plateaux in the south to the Mediterranean Sea in the north. In the east, it is limited by the Kebir-West and the Seybouse basins and on the west by the Soummam basin. We can define three main parts: (a) an upper basin corresponding to Rhumel and Boumerzoug wadis up to their confluence, upstream of Constantine, (b) a middle basin: formed by the Endja wadi and lower Rhumel wadi, downstream of Constantine, to their

junction at Sidi Mérouane, and (c) a lower basin: corresponding to the Rhumel-Kebir wadi, downstream of the junction Endja-Rhumel (Mebarki, 1982).

The Rhumel hydrosystem is 4200 km long and is represented by 43 wadis; the most representative are the Rhumel and Endja wadis, which respectively drain an area of 5315 km² and 2160 km² and whose confluence gives rise to Rhumel-Kebir (Fig. 1d).

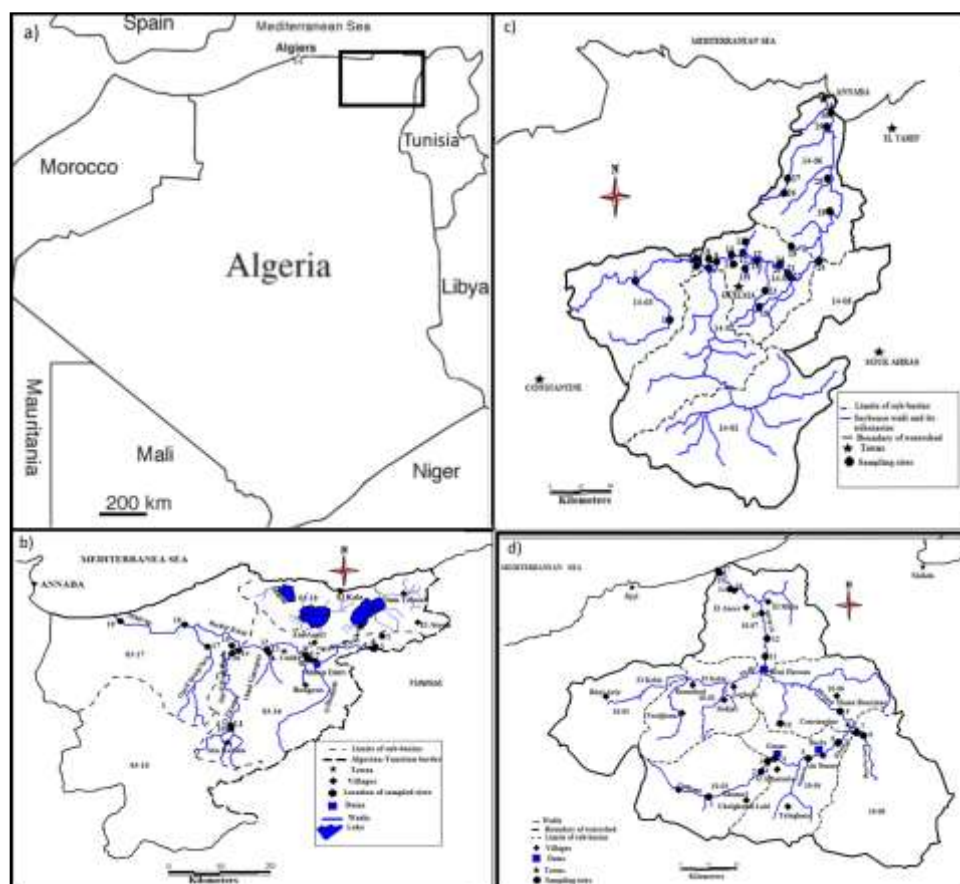


Figure 1.— Location map of Algeria and watershed of Kebir East, Seybouse and Rhumel.

The climate overlaps between three bioclimatic zones from semi-arid to sub-humid and humid. The rainfall decreases from north to south and varies from 1000 mm in the north to 318 mm in the south (ABHCSM, 1999).

The most harmful effluents are located within 20 km of Constantine, near Ain Smara where car industries, textiles, milk, detergent and cement plants are located. These untreated effluents are discharged into the Rhumel and its main tributary, the Boumerzoug (Mebarki, 1982).

SAMPLING STRATEGY

Site selection was based on site accessibility, altitude and variability of habitats. Some stations were sampled regularly while others were occasionally visited (Tabs. I, II, III; Figs. 1 a,b,c). Monitoring of the odonatofauna was carried out at Seybouse, during the years 2007. In 2011, both Kebir-East and Rhumel were sampled whereas in 2013 the survey was limited to the Rhumel. Monthly sampling was carried out from March to November; a span which overlaps with the flight period of Algerian Odonata (Samraoui & Corbet 2000b).

Sampling was standardized at all sites and lasted one hour: adults Odonata were caught using a butterfly net along both banks of the sampled wadi. Adult behavior was noted (territoriality, mating, oviposition) and other information collected to identify and estimate anthropogenic factors (industrial units, water extraction, nature of waste, infrastructure, land use). Slope was measured, using digital field models with a resolution of thirty meters.

TABLE I

List of geographical location of sampled sites at Kebir-East (W) and its main tributaries (T)

Name	Status	Station code	Station number	Latitude	Longitude	Altitude (m)	Hydroperiod
Oued El Hout*	T	O El H	1	36° 49. 424'N	8° 29. 298' E	346.5	Temporary
Leben	T	Leb	2	36° 48. 226' N	8° 32.132' E	111.5	Temporary
Ain Dem*	T	Ain Dem	3	36°46.979' N	8° 31.246' E	84.9	Temporary
Kebir Raml Souk 1	W	KRS 1	4	36° 46 .654' N	8° 31.053' E	82.3	Permanent
Kebir Raml Souk 2	W	KRS 2	5	36° 46 .758' N	8° 30.841' E	86.7	Permanent
Mexa1	W	Mexa1	6	36° 45.372' N	8° 22.748' E	33.1	Permanent
Mexa2	T	Mexa2	7	36° 45.269'N	8° 23.313' E	25.2	Temporary
Deep Kebir	W	K Pro	8	36° 46 .025' N	8°21.922' E	41	Permanent
Oued Ben Sebti *	T	O Ben Se	9	36° 45.542' N	8°22.252' E	36.4	Temporary
Near Bouhaloufa *	T	N Bouhal	10	36° 37.141' N	8° 12.266' E	143.6	Permanent
Oued Bouhaloufa	T	O Bouhal	11	36° 37.541' N	8° 12.147' E	127.5	Temporary
Kebir near Guergour *	W	ks Guerg	12	36° 46.653' N	8°16. 886' E	28	Permanent
Guergour	T	Guergour	13	36° 46;284' N	8° 17.220' E	24.9	Permanent
Dardan*	T	Dardan	14	36° 46.523' N	8° 13.212' E	8.3	Temporary
Oued Oum Saad *	T	Oum Saad	15	36° 47.154' N	8° 12.312' E	17.6	Temporary
Kef Belechehab *	T	Kef Bele	16	36° 46.166' N	8° 12.312' E	18	Temporary
Boulathane	T	Boulathane	17	36° 47.009' N	8°09.105' E	4.7	Temporary
K Sebaa	W	Sebaa	18	36° 49.704' N	8° 06.112' E	8.6	Permanent
Kebir Mafragh	w	K Maf	19	36°50.178' N	7° 57.621' E	1	Permanent

*occasionally sampled sites

TABLE II

List of geographical location of samples sites at Seybouse (W) and its main tributaries (T)

Name	Status	Station code	Station number	Latitude	Longitude	Altitude (m)	Hydroperiod
Oued Zenati	T	O Zen	1	N36°18.208'	E 7°09.862'	622.2	Permanent
Oued Sabath	T	O Sab	2	N36°24.703'	E 7°02.461'	620.6	Permanent
Chedakha*	T	Ched	3	N36°27.040'	E 7°16.002'	329.2	Permanent
Hammam Skhoun*	T	H Skh	4	N36°27.682'	E 7°16.274'	286.2	Permanent
Bouhamdane1 *	T	Bouha 1	5	N36°27.883'	E 7°16.274'	260.5	Permanent
Bouhamdane2 *	T	Bouha 2	6	N36°27.996'	E 7°15.802'	268.3	Permanent
Medjaz Amar	T	Mej2	7	N36°26.597'	E 7°18.619'	243.1	Permanent
Seybouse (Djebala)*	W	SeyDje	8	N36°28.098'	E 7°18.619'	157.3	Permanent
Seybouse (Bentabouche cliff)*	W	Sey Ben	9	N36°27.659'	E 7°20.352'	230.6	Permanent
Bridge(Fedjouj- Skikda)*	W	SeyFS	10	N36°28.687'	E 7°23.806'	208.7	Permanent
Bendjerrah (Slope)*	T	Benj	11	N36°27.180'	E 7°24.238'	435.3	Temporary
Seybouse (Guelma-Annaba)*	W	Seypga	12	N36°29.094'	E 7°26.312'	190.2	Permanent
Bradaa 2*	T	Brad2	13	N36°30.841'	E 7°26.979'	259.7	Permanent
Seybouse*	W	SeyBou	14	N36°27.108'	E 7°34.349'	144.5	Permanent
Oued El Maiz*	T	OEIM	15	N36°26.641'	E 7°26.742'	305.4	Permanent
Zimba (Belkhir)	T	Zim	16	N36°27.816'	E 7°29.220'	205.3	Permanent
Boumahra*	W	BouVil	17	N36°27.781'	E 7°29.212'	190.2	Temporary
Oued Hlia (Downstream)*	T	O Hlia	18	N36°20.116'	E 7°29.661'	433.6	Permanent
Seybouse (Laayaycha)*	W	Seylaa	19	N36°26.913'	E 7°36.908'	152.3	Permanent
Boumahra (station)	W	SeyBou	20	N36°26.999'	E 7°34.225'	135.4	Permanent
Seybouse (Farm Chekal, Nador)*	W	SeyNad	21	N36°25.538'	E 7°36.112'	125.5	Permanent
Oued Hlia (near Nador)*	T	Ohlia	22	N36°24.792'	E 7°36.672'	132.6	Permanent
Oued Boussera (Khezara)*	T	Obous	23	N36°22.717'	E 7°30.970'	306.7	Permanent
Oued El Maieh (Boucheghouf)	T	Omaleh	24	N36°27.494'	E 7°42.989'	97.3	Permanent
Seybouse Boukamouza	W	SeyBou	25	N36°35.555'	E 7°45.615'	56.4	Permanent
Ressoul (El Barda)*	T	Ressoul	26	N36°38.717'	E 7°35.479'	102	Permanent
Meboudja (El Barda)	T	MebBer	27	N36°41.083'	E 7°36.322'	57.8	Permanent
Seybouse (Drean)	W	SeyDr	28	N36°40.999'	E 7°45.077'	20.7	Permanent
Meboudja (industrial zone)*	W	MebZI	29	N36°49.440'	E 7°45.077'	6.6	Permanent
El Aalalick *	W	Alalick	30	N36°51.792'	E 7°46.066'	4.5	Permanent
Seybouse (Sidi Salem)*	W	SeySi	31	N36°51.687'	E 7°46.082'	1.3	Permanent

*occasionally sampled sites

PHYSICO-CHEMICAL DATA

Measures of pH, water temperature, electrical conductivity, salinity, dissolved oxygen and water velocity as well as observations of vegetation types, percentage of shading and vegetation cover, water depth and wadi width were recorded for each station.

DATA ANALYSIS

Preliminary examination of the data sets pointed to shortcomings of the adopted sampling and suggested that the analysis of a combined data set would be problematic. Indeed, logistic difficulties made it difficult to sample uniformly (years, altitude, etc.) all three wadis as initially planned, making it difficult to combine the three data sets. Thus, separate multivariate and univariate analyses of each wadi were performed. After data exploration to check for outliers and collinearity, we started with dimensionality reduction using a PCA (environmental data set). A reduced set of abiotic variables was then used as supplementary variables in a second PCA based on the faunistic data set (absence/presence). PCAs were performed with the FactoMineR package (Lê *et al.*, 2008). A co-inertia analysis (Doledec & Chessel, 1994), involving both faunistic and environmental data sets, was also performed for each wadi and followed by a generalized linear model (GLM) to identify environmental determinants (among the reduced set of environmental variables conveyed

by the PCAs) of species richness: we modelled species richness using a Poisson GLM with a logit link function and poisson error distribution. Backward selection, based on the Aikaike information criterion (AIC) was performed to select the most parsimonious model (Burham & Anderson, 2002). All statistical analyses were carried out using R (R Development Core Team 2016).

TABLE III
List of geographical location of samples sites at Rhumel (W) and its main tributaries (T)

Name	Status	Code station	Station number	Latitude	Longitude	Altitude (m)	Hydroperiod
Rhumel at Tadjent	W	Rh Tadj	1	36° 07.137' N	6° 02.809' E	814.6	Permanent
Rh HG upstream	W	Rh HG Am	2	36° 12.141' N	6° 15.417' E	722.3	Permanent
RH HG downstream	W	RH HG Aval	3	36° 14.174' N	6° 16.910' E	677.2	Permanent
RH O Athmania	W	RH Athm	4	36° 14.747' N	6° 18.617' E	696.7	Permanent
RH Cascade	W	RH Cascade	5	36° 14.705' N	6° 26.662' E	628.9	Permanent
RH Ain Smara	W	RH Smara	6	36° 17.839' N	6° 33.840' E	564.3	Permanent
Boumerzoug (Chaabat Rsas)	T	Boum Rsas	7	36° 19.835' N	6° 38.142' E	737	Permanent
Boumerzoug (Sisaoui)	W	Boum Sisa	8	36° 19.224' N	6° 39.790' E	554.2	Permanent
RH Hama Bouziane	W	RH Bouziane	9	36° 23.984' N	6° 34.290' E	338.7	Permanent
Oued El Koton*	T	O El Kot	17	36° 21.021' N	6° 18.305' E	809	Temporaire
Chegara	W	Chegara	10	36° 35.120' N	6° 16.570' E	66.6	Permanent
Kebir at Rafraf	W	Keb raf	11	36° 38.581' N	6° 16.940' E	33.4	Permanent
Kebir, bridge of El Milia*	W	Keb Milia	12	36° 43.600' N	6° 15.714' E	30	Permanent
Djemaa	T	Djemaa	13	36° 48.433' N	6° 08.085' E	19.3	Temporaire
Boumaazouz	T	Boumaazouz	14	36° 51.670' N	6° 05.963' E	14.1	Temporaire
Irdjana	W	Irdjana	15	36° 48.254' N	6° 09.341' E	9.2	Permanent
Kebir Djenah	W	Keb Dje	16	36° 52.220' N	6° 05.312' E	0.7	Permanent

* Occasionally sampled sites

RESULTS

Overall, the present survey yielded 40 species with 34, 33, and 32 species recorded at Kebir-East, Seybouse, and Rhumel wadis, respectively (Tab. IV). Of these, 17 species were Zygopterans while 23 were Anisopterans. The list included four species considered threatened by the IUCN North African Red List (Samraoui *et al.*, 2010): *Urothemis edwardsii* (CR), *Calopteryx exul* (EN), *Acisoma inflatum* (EN) and *Gomphus lucasii* (EN). Five species (*Lestes numidicus*, *Enallagma deserti*, *Platycnemis subdilatata*, *C. exul* and *G. lucasii*) are Maghrebien endemics whereas two (*U. edwardsii* and *A. inflatum*) are Afrotropical relicts with reduced populations (Samraoui *et al.*, 1993).

DATA ANALYSES

Wadi Kebir-East

The first two principal components of the PCA, based on physicochemical variables accounted for 62.4 % of the total variance (Fig. 2a). The first component (37.6 % of the inertia) was associated with stations of high water conductivity, salinity, water depth and wadi width opposed to stations characterized by high current velocity. The second component (24.8 % of the inertia) was associated with long hydroperiods and opposes stations with high pH and dissolved oxygen levels. Species richness added as a supplementary variables appeared to be negatively correlated to the hydroperiod (Fig. 2a).

In contrast, the first two principal components of the PCA based on species accounted for only 36.6 % of the total variation (Fig. 2b). The first component (19.1 % of the inertia) separated species (*Platycnemis subdilatata*, *Lestes virens*, *Orthetrum coerulescens*, *Aeshna mixta*, *Ischnura pumilio* and *Ceriagrion tenellum*) associated with shade from those (*Onychogomphus costae*, *Gomphus lucasii*, *Erythromma viridulum*, *Diplacodes lefebvrei* and *Crocothemis erythraea*) associated with more exposed stations. The second component (17.7 % of the inertia) seemed to separate two more groups on the basis of altitude: *C. caerulescens*, *Trithemis kirbyi* and *T. arteriosa* were more frequent at higher altitudes whereas *Brachythemis impartita*, *Lestes barbarus*, *Coenagrion mercuriale* and *Sympetrum meridionale* were more frequent at lower altitudes stations with high water conductivity.

TABLE IV

Status and species richness in the three study sites

	Kebir-East	Seybouse	Rhumel	IUCN Status
<i>Calopteryx exul</i> Selys, 1853	No record	7, 10	No record	EN
<i>C. haemorrhoidalis</i> (Vander Linden, 1825)	2	3, 11, 13, 15-17, 19	14	LC
<i>Sympecma fusca</i> (Vander Linden, 1820)	No record	Standing waters	6	LC
<i>Lesies barbarus</i> (Fabricius, 1798)	4, 7, 14, 16, 17, 19	Standing waters	13-15	LC
<i>L. numidicus</i> Samraoui, Weekers & Dumont, 2003	7, 18	20	No record	DD
<i>L. virens</i> (Charpentier, 1825)	5, 6, 10, 11, 14, 16, 18, 19	15	14	LC
<i>Chalcolestes viridis</i> (Vander Linden, 1825)	2-8, 13, 16, 17, 19	2, 13, 20	4-7, 10, 12, 14-16	LC
<i>Platycnemis subdilata</i> Selys, 1849	2, 5-14, 17-19	1-3, 7, 9, 10, 12, 13, 16-21, 23, 27, 28	3, 5-7, 9-11, 13	LC
<i>Ischnura graellsii</i> (Rambur, 1842)	4-9, 11-19	1-3, 5-9, 13, 16, 17, 20, 25-28, 31	1-3, 5-9, 11, 13-17	LC
<i>I. pumilio</i> (Charpentier, 1825)	11	10	1, 2, 7, 8, 13	LC
<i>Coenagrion caeruleum</i> (Fonscolombe, 1838)	2, 6	1, 2, 5-8, 19, 20	1-3, 5-9	LC
<i>C. mercuriale</i> (Charpentier, 1840)	7, 8, 13, 15, 17-19	No record	2, 5-9, 13, 17	EN
<i>C. puella</i> (Linnaeus, 1758)	No record	Standing waters	No record	LC
<i>Enallagma deserti</i> (Selys, 1871)	No record	Standing waters	7	LC
<i>Erythronia lindesii</i> (Selys, 1840)	2-7, 9, 11, 15, 17, 18	1, 6, 8-10, 12, 20	1-3, 5-7, 12, 15-17	LC
<i>E. viridulum</i> (Charpentier, 1840)	2-5	No record	No record	LC
<i>Ceragrion tenellum</i> (de Villers, 1789)	2, 6, 13, 18	16, 17, 19	15	LC
<i>Gomphus lucasii</i> Selys, 1849	4-8, 17	10, 13, 18	12-14	VU
<i>Onychogomphus costae</i> Selys, 1885	4, 5, 7	12, 21	12	NT
<i>O. forcipatus unguiculatus</i> (Vander Linden, 1823)	No record	No record	13, 16	LC
<i>Paragomphus genei</i> (Selys, 1841)	4	No record	16	LC
<i>Aeshna mixta</i> Latreille, 1805	1, 6, 9, 11, 16, 18	13, 25	13, 15, 17	LC
<i>Anax imperator</i> Leach, 1815	1, 5-7, 13	2, 13, 15, 19, 20, 24, 30	3-5	LC
<i>A. parthenope</i> (Selys, 1839)	1, 7, 13	Standing waters	2	LC
<i>Orthetrum cancellatum</i> (Linnaeus, 1758)	No record	30	3	LC
<i>O. chrysostigma</i> (Burmeister, 1839)	4-11, 13, 14, 16, 17	1-4, 7-9, 11-13, 15, 16, 18, 24, 26, 27	3, 4, 8, 9, 11, 12, 14-17	LC
<i>O. coerulescens</i> (Fabricius, 1798)	3, 6, 8, 9, 13, 15, 17	2-4, 11, 16, 19, 27, 28	2-4, 9, 11, 15	LC
<i>O. nitidinerve</i> (Selys, 1841)	6, 7	2, 3, 13, 19, 20	1-6, 9	LC
<i>O. trinacria</i> (Selys, 1841)	7	No record	No record	LC
<i>Acisoma inflatum</i> Selys, 1889	6	No record	No record	EN
<i>Brachythemis imparita</i> (Karsch, 1890)	4, 7, 17, 18	28, 31	2, 15-17	LC
<i>Crocothemis erythraea</i> (Brullé, 1832)	1-7, 11, 13, 14, 16, 19	1, 3, 5, 6, 8, 16, 18-20, 24, 26, 28, 30, 31	1-7, 9, 14-16	LC
<i>Diplacodes lefebvrei</i> (Rambur, 1842)	1-7, 19	26, 28, 30, 31	3, 15	LC
<i>Sympetrum fonscolombii</i> (Selys, 1840)	4, 5, 14, 17-19	1, 2, 10, 13, 20, 27, 28, 30, 31	1-4, 6-9, 12, 15-17	LC
<i>S. meridionale</i> (Selys, 1841)	1, 4, 8, 16-19	4, 5, 13, 25, 27, 28	9, 15, 17	LC
<i>S. striolatum</i> (Charpentier, 1840)	1, 3-6, 8, 9, 11, 13, 16-18	4, 10, 20, 26	4-6, 11, 13, 15-17	LC
<i>Trithemis annulata</i> (Palisot de Beauvois, 1807)	2, 4-8, 13, 17-19	1, 16, 26-28	3-6, 9, 11, 15-17	LC
<i>T. arteriosa</i> (Burmeister, 1839)	5, 6, 13, 14	19, 26, 27	3	LC
<i>T. kirbyi</i> Selys, 1891	2, 4, 6	2, 11, 13, 18, 26	No record	LC
<i>Urothemis edwardsii</i> (Selys, 1849)	5	No record	No record	CR

Results of the CIA (Fig. 3) illustrated the projection of the stations (K1-K11). Stations with high shade and current velocity were separated from other stations with high pH and dissolved oxygen levels. The second axis was mainly an altitudinal gradient separating stream stations from those low-altitudes stations characterized by wide width and high water depth. The RV-coefficient, an index of the correlation between the two matrices (species and environmental variables), was 0.58. A permutation test explored the strength of the relationship between the two matrices but indicated no significance, presumably because of the limited sample size. The CIA organized the Odonata assemblages of Kebir-East along both shade/current velocity and altitudinal axes.

Univariate modelling using Poisson GLM and species richness as a response variable indicated that the hydroperiod was the most significant covariate (Tab. V, Fig. 8a) but the “best” model was made up of both hydroperiod and altitude. However, this latter covariate was only marginally significant ($p = 0.099$) suggesting that both models were essentially equivalent. Sites with longer hydroperiods held fewer species of Odonata.

Wadi Seybouse

The first two principal components of PCA, based on the environmental data set, accounted for 65.6 % of the total variation (Fig. 4a). The first component (41.3 %) separated high altitudes stations from lower altitudes ones characterized by high conductivity and current velocity. The second component (24.3 % of the inertia) opposed altitude and conductivity to temperature, pH, and dissolved oxygen levels. Species richness, used as a supplementary variable, was opposed to current velocity, slope, high conductivity and salinity.

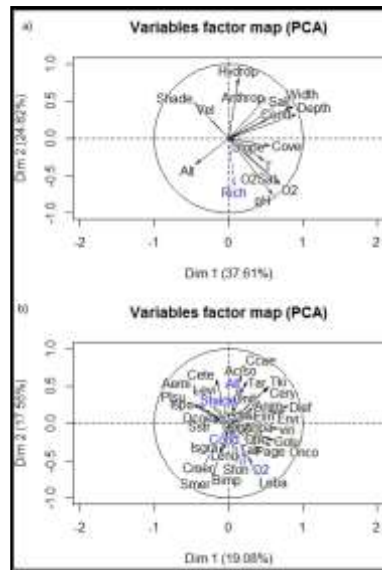


Figure 2.— Correlations between measured environmental variables of Wadi Kebir-East mapped onto the first two principal components of PCA. Species richness was used as a supplementary variable.

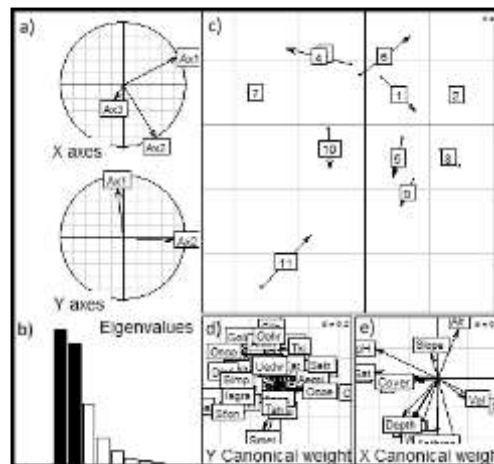


Figure 3.— Results of the co-inertia analysis (CIA) performed between recorded Odonata (coded as presence/absence) and measured environmental variables at Wadi Kebir-East. A) Components of the standardized principal component analysis of the environmental data set projected onto the co-inertia axes (top) and components of the centred principal component analysis of the odonatological data set projected onto the co-inertia axes (bottom). B) Distribution of eigenvalues of CIA. C) F1 x F2 factorial plane of CIA with arrows linking stations according to physico-chemical variables (base of arrows) and Odonata (end of arrows). D) Distribution of species on the F1 x F2 factorial plan of CIA. E) Distribution of environmental descriptors on the F1 x F2 factorial plan of CIA.

Likewise, the first two principal component of PCA, based the faunistic data set, accounted for only 42.9% of the total variation (Fig. 4b). The first component (23.9 % of the inertia) isolated species associated with high altitudes (*Orthetrum nitidinerve*, *Chalcolestes viridis*, *Anax imperator* and *Trithemis kirbyi*) from species associated with lower altitudes and high water conductivity

(*Brachythemis impartita*, *Diplacodes lefebvreii*, *Trithemis annulata* and *Orthetum coerulescens*). The second component (19.0 % of the inertia) separated species recorded at stations with higher water temperatures and dissolved oxygen (*Gomphus lucasii*, *Sympetrum meridionale*, *Aeshna mixta* and *Orthetrum chrysostigma*) from a group which included *Erythromma lindenii*, *Crocothemis erythraea*, *Sympetrum striolatum*, *Lestes numidicus* and *Coenagrion caerulescens*.

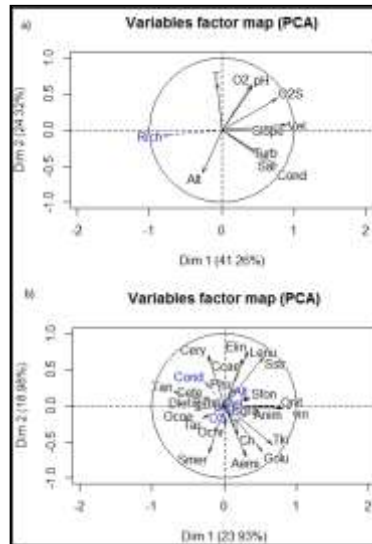


Figure 4.— Correlations between measured environmental variables of Wadi Seybouse mapped onto the first two principal components of PCA. Species richness was used as a supplementary variable.

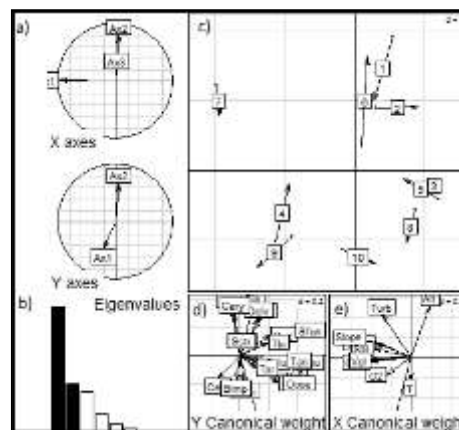


Figure 5.— Results of the co-inertia analysis (CIA) performed between recorded Odonata (coded as presence/absence) and measured environmental variables at Wadi Seybouse. A) Components of the standardized principal component analysis of the environmental data set projected onto the co-inertia axes (top) and components of the centred principal component analysis of the odonatological data set projected onto the co-inertia axes (bottom). B) Distribution of eigenvalues of CIA. C) F1 x F2 factorial plane of CIA with arrows linking stations according to physico-chemical variables (base of arrows) and Odonata (end of arrows). D) Distribution of species on the F1 x F2 factorial plan of CIA. E) Distribution of environmental descriptors on the F1x F2 factorial plan of CIA.

Results of the CIA (Fig. 5) illustrated the projection of the stations (S1-S10). Stations with high current velocity, conductivity and dissolved oxygen levels dominated the first axis. The second axis represented essentially an altitudinal gradient separating stream stations from those low-altitudes stations characterized by high pH and water temperature. The RV-coefficient, an index of the correlation between the two matrices (species and environmental variables), was 0.51 but, likewise, the permutation test did not show a significant relationship for probably identical reasons (reduced sample size). The Odonata assemblages at Wadi Seybouse were thus organized by CIA along both conductivity/velocity/dissolved oxygen and altitudinal axes.

TABLE V
Parameters estimates for the poisson GLM analysis of Odonata species richness at three Algerian wadis

	Estimate	95% CI	z value	p
<i>Kebir-East</i>				
intercept	44.451	19.685-100.376	9.13	<0.0001
Hydroperiod	0.996	0.994-0.999	-2.714	0.0067
Altitude	0.996	0.992-1.001	-1.652	0.0986
<i>Seybouse</i>				
intercept	30.567	10.999-85.0209	6.552	<0.0001
O2	0.877	0.752-1.023	-1.667	0.096
Conductivity	0.999	0.999-0.999	-2.167	0.03
<i>Rhumel</i>				
intercept	1.699	0.189-15.264	0.474	0.636
Temperature	1.075	0.981-1.177	1.554	0.12

Univariate modelling using Poisson GLM and species richness as a response variable indicated that conductivity was the most significant covariate (Tab. V, Fig. 8b) but the best model was made up of both conductivity and dissolved oxygen. However, this latter covariate was again only marginally significant ($p = 0.096$) and the AIC can safely be overruled by retaining a simplified model. Species richness at Seybouse decreased with increasing values of water conductivity.

Wadi Rhumel

The first two principal components of PCA, based on the environmental data set, accounted for 58.0 % of the total variation (Fig. 6a). The first component (34.2 % of the inertia) separated stations with high water conductivity, salinity and hard water from stations with well oxygenated water. The second principal component (23.9 % of the inertia) characterized stations with high pollution loads. In contrast, the first two principal components of PCA, based on the environmental data set, accounted for only 35.0 % of the total variation (Fig. 6b). The first principal component (18.49 % of the inertia) isolated species like *Coenagrion caerulescens*, *Orthetrum nitidinerve*, *Trithemis arteriosa*, *Diplacodes lefebvreii*, *Orthetrum cancellatum*, *Platycnemis subdilatata* and *Crocothemis erythraea*) associated with higher altitudes and having high water conductivity and water hardness from *Paragomphus genei*, *Gomphus lucasii*, *Onychogomphus costae*, *Brachythemis impartita*, *Sympetrum striolatum* and *Orthetrum chrysostigma* associated with lower altitude stations exhibiting higher dissolved oxygen content. The second component (16.6 % of the inertia) isolated an assemblage which included *Coenagrion mercuriale*, *Ischnura pumilio*, *I. graellsii*, *Enallagma deserti* and *Sympetrum fonscolombii* which were associated with waters subjected to a high organic pollution load from another group composed of *Trithemis annulata*, *Orthetrum chrysostigma* and *Anax imperator*, presumably more sensitive to such pollution.

Projection of the stations (R1-R13) by the CIA indicated that the lower reaches of Wadi Rhumel were characterized on the first axis by high water depth, oxygen levels, temperature and wadi width, distinct from stations with higher water conductivity (Fig. 7). The second axis represented a pollution gradient. The RV-coefficient was 0.60 but, likewise, the permutation test

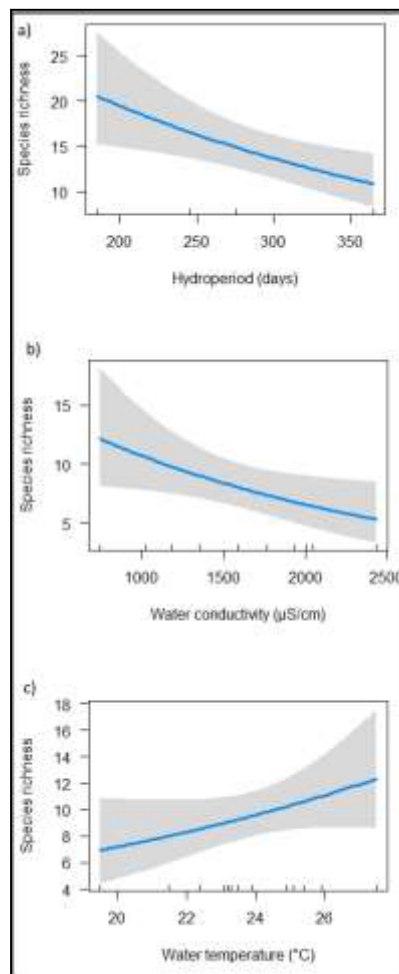


Figure 8.— Predicted species richness plotted against covariates at (a) Kebir-East, (b) Seybouse, (c) Rhumel wadis. Shaded area shows 95 % confidence intervals for the mean.

Univariate modelling of the species richness as a response variable by a Poisson GLM indicated that temperature was the most important covariate, albeit in a marginal way (Tab. V, Fig. 8c). The number of recorded species rose as water temperature increased.

DISCUSSION

In total, 40 species were recorded in our survey with 33, 34 and 32 species for Seybouse, Kebir-East and Rhumel, respectively. It should be noted that all species were sampled in running waters with the exception of *Coenagrion puella*, recorded in standing waters at the Seybouse watershed (Tab. IV). The recorded species represented 60 % of the Algerian odonatofauna (Samraoui & Menai, 1999); 26 species are common to the three wadis. The list includes endemic and threatened species that are indexed in the IUCN North African Red List (Samraoui *et al.*, 2010) (Tab. II).

Two species, *Coenagrion caerulescens* and *Trithemis kirbyi* are new to Numidia (Samraoui & Corbet, 2000a). The latter species has extended its range northwards over the last decades (Boudot

et al., 2009). Another species, *Coenagrion mercuriale*, previously considered as rare, was found to be relatively abundant and common in fast flowing wadis. The species is classified as endangered in the IUCN North African Red List (Samraoui *et al.*, 2010) and Near Threatened in the IUCN Mediterranean Red List (Riservato *et al.*, 2009).

Among the Gomphidae, we recorded four species: *Gomphus lucasii*, *Onychogomphus costae*, *O. forcipatus unguiculatus* and *Paragomphus genei*. Endemic to the Maghreb, *Gomphus lucasii*, found in all three wadis, is classified as vulnerable (VU) in the IUCN North African Red list (Samraoui *et al.*, 2010), the IUCN Mediterranean Red List (Riservato *et al.*, 2009) and the global Red List.

In contrast, *Onychogomphus forcipatus unguiculatus*, assessed as Least Concern in the IUCN North African Red List (Samraoui *et al.*, 2010) and the IUCN Mediterranean Red List (Riservato *et al.*, 2009), was considered rare by Samraoui & Corbet (2000a) and our study seemed to confirm this precarious status for the eastern part of the country. Elsewhere in Algeria, the species appears to be more common (Hamzaoui *et al.*, 2015).

Some species were confined to one wadi (*Calopteryx exul*) or were recorded infrequently on only one site (*Acisoma inflatum*, *Urothemis edwardsii* and *Lestes numidicus*). Endemic to the Maghreb, *Calopteryx exul* is now known in Algeria from only the Seybouse wadi. This species was last recorded by Martin (1910) before its rediscovery at locality S16 during the present study. Following this record, more localities were found by various other investigators but were all limited to the Seybouse (Khelifa *et al.*, 2011; unpublished). *C. exul* is also classified as endangered in the IUCN Mediterranean Red List (Riservato *et al.*, 2009) and the global Red List.

Records of two relict Afrotropical species *Acisoma inflatum* and *Urothemis edwardsii* at the Kebir-East were also made. The presence of the type population of *U. edwardsii* which was for two decades confined to a single site, Lac Bleu, following the demise of another site, Lac Noir (Bélair & Samraoui, 1994), is remarkable. This record is congruent with other observations made at the same time confirming a sudden expansion of the species in the region (Samraoui, submitted). In the Mediterranean region, *A. inflatum* and *U. edwardsii* are unique to the El Kala region and neighbouring wetlands (Samraoui *et al.*, 1993). Both threatened species are listed in the IUCN Mediterranean Red List (Riservato *et al.*, 2009) and the IUCN North African Red List (Samraoui *et al.*, 2010).

Two males of another North African endemic, *Lestes numidicus*, were caught. The species is classified as data deficient (DD) in the IUCN North African Red List (Samraoui *et al.*, 2010), the IUCN Mediterranean Red List (Riservato *et al.*, 2009) and the global Red List. In Algeria, the study of the complex "*Lestes virens*" revealed the presence within the complex of two taxa that can be separated by rDNA analysis, adult coloration and reproductive period (Samraoui *et al.*, 2003; Samraoui, 2009).

The ecological goods and services provided by Algerian wadis in the arid North African landscape have been eroded over the past decades as local needs for freshwater rose sharply. Like many Mediterranean rivers, Algerian wadis have been subjected to intensive anthropogenic stressors through dam construction and water abstraction (Muñoz & Prat, 1996; Jones, 2011), afforestation with widespread use of exotic pine (Bellot *et al.*, 2004), *Acacia* and *Eucalyptus* plantations (Tererai *et al.*, 2013), invasion of alien plants like Giant Reed (Quinn & Holt, 2008), mining activities (Carvalho *et al.*, 2014; Ferreras-Romero *et al.*, 2009), channel straightening (Elosegi *et al.*, 2010), industrial pollution (Gonzalo & Camargo, 2013) and agricultural runoff (Thiere & Schulz, 2004). These anthropogenic factors can profoundly alter the natural flow regimes and degrade the water quality of wadis, thus impacting negatively the native biodiversity and ecological processes.

Our results are congruent with previous research indicating that dragonflies and damselflies (both adults and larvae) respond to physical habitat conditions (Corbet, 1999) but differed to some extent. Species richness at Kebir-East was negatively correlated with the hydroperiod. Drought

and low flows are known to influence macroinvertebrates (Wood & Petts, 1994). When looking for suitable ovipositing sites, adult Odonates may use visual cues and respond to the horizontal polarization of light reflected from the water surface (Corbet, 1999) and avoid dry stretches of rivers and streams (Hardersen, 2008). However, stations with longer hydroperiods at Kebir-East are correlated with high degree of anthropogenic impacts and are thus shunned by Odonata.

Another important environmental parameter in determining the distribution of Odonata is altitude (Jacobsen, 2003; Harabis & Dolny, 2010). Alongside species associated with the semi-arid Hauts Plateaux like *Coenagrion caerulescens* and *Orthetrum nitidulerve*, we recorded two species of *Trithemis*: *T. arteriosa* and *T. kirbyi*, the latter being known to have expanded its range northward over a recent past (Boudot & Kalkman 2015; Brouwer 2015). Their presence at high altitude in coastal areas may be symptomatic of an altitudinal shift similar to the more obvious latitudinal northward shift (Hickling *et al.*, 2005).

Riparian site selection by breeding Odonates is influenced by the extent of shade and the structure of understory vegetation (Remsburg *et al.*, 2008) thus afforestation may impact negatively on river dynamics and species diversity (Kinzig & Samways, 2000). Shade-seeking species like *Ceragrion tenellum* and *Orthetrum coerulescens* are typically associated with dense riparian vegetation which may itself positively influence species richness (Schindler *et al.*, 2010). *Aeshna mixta*, during its aestivation period may also be associated in dense, shaded forests (Samraoui *et al.*, 1998).

Depending on the extent of the canopy, shaded water and riparian areas have probably lower air and water temperature as these two variables are closely related although this relationship may be non-linear (Morrill *et al.*, 2005). Thus, it comes as no surprise that species richness may be driven by temperature as was found at the Rhumel. Temperature is known to set constraints on the distribution (Corbet, 1999; Oertli, 2010), behaviour (De Marco *et al.*, 2005; Reinhardt, 2006; Corbet & May, 2008) and ecology of Odonata (Lutz, 1968; May, 1976; Corbet, 1999).

Among the heliothermic species recorded at sunlit stations, the Gomphidae were represented by *Gomphus lucasii*, and *Onychogomphus costae*. These two species were followed by another gomphid, *Paragomphus genei*, in well-oxygenated sites. Gomphids are sensitive species that shun water pollution (Ferrerias-Romero *et al.*, 2009). Indeed, sewage effluents negatively affect the abundance and diversity of both adult and larval stages of Odonates (Watson *et al.*, 1982) and low pH and high levels of water conductivity may impact adversely Odonates species richness (Rychla *et al.*, 2011). Our results for Wadi Seybouse concur with these findings. The presence of two species of *Ischnura*: *I. graellsii* and *I. pumilio* is unsurprising as species of *Ischnura* have been shown to be quite tolerant of water pollution (Ferrerias-Romero *et al.*, 2009; Campos *et al.*, 2014).

Our preliminary findings suggest that Mediterranean Odonata communities respond to environmental factors and that assessments of their species richness may offer a simple and rapid tool to monitor the ecological integrity of lotic systems. However, there is a real need to increase our knowledge of variations of this index across different scales of space and time.

CONCLUSION

Our investigations of three wadis, Kebir-East, Seybouse, and Rhumel, led to 40 species being recorded. Among these, *Coenagrion coerulescens* and *Trithemis kirbyi* are new to Numidia. The threatened Maghrebian endemic, *Calopteryx exul*, was first recorded in the Seybouse during this study, and confirmed as probably extinct in its type locality, the Rhumel. Our results also confirmed the recent expansion of the threatened relict species *Urothemis edwardsii* which was previously confined to a single site. Data analysis indicated a good congruence between the faunistic and environmental data sets while the exploration of species richness as a surrogate for Odonata diversity, suggested that this ecological index may vary differentially being associated,

according to the wadi studied, with hydroperiod, water conductivity and water temperature. Odonata assemblages as reflected by species richness seem to be strongly shaped by physical factors.

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